

CLIMATE CHANGE AND COMMON RESOURCES

PATHWAYS TO ARTS & HUMANITIES SUMMER SCHOLARS PROGRAM

Instructor: Tudor Schlanger

Summer 2026

REVIEW: CORRELATION VS. CAUSALITY

Question

A study finds that cities with more firefighters also have more fire damage.

A local politician concludes: “There are too many firefighters, they get in each other’s way and do more bad than good. They **cause** more fire damage! So we should hire fewer firefighters!”

What is wrong with this reasoning? You can choose one or more options.

- ☐ A. Nothing, it’s a true causal statement
- ☐ B. Reverse causality
- ☐ C. Hidden third factor (Confounder)

REVIEW: CORRELATION VS. CAUSALITY — ANSWER

Answer

The correct answers are:

B. Reverse causality: More fire damage causes cities to hire more firefighters — the causation runs the opposite direction.

C. Hidden third factor: Bigger cities have both more fires *and* more firefighters. City size is a confounder.

Seeing two things move together (correlation) does not tell you which one causes the other — or whether something else drives both.

OUTLINE

1. The Candy Problem

2. Carbon Tax Policy Discussion

3. Recap

YALE POLICY 101: KIDS DESERVE CANDY!

Rules:

1. Line up
2. Walk up to the candy bowl
3. Take as many pieces as you want (for you, your family, friends...)
4. No limits, no penalties

Seriously — no limits.



YALE POLICY 101: CONSEQUENCES

YALE DAILY NEWS
New Haven, CT Summer 2026

**CANDY CRISIS:
Policy 101 Depletes
Yale's Candy Stocks**

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YALE HERALD
New Haven, CT Summer 2026

**UNFAIR DISTRIBUTION:
Not Everyone Got Candy
Under Policy 101**

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Yale Policy 102: KIDS DESERVE **ONE** CANDY!!

New rules:

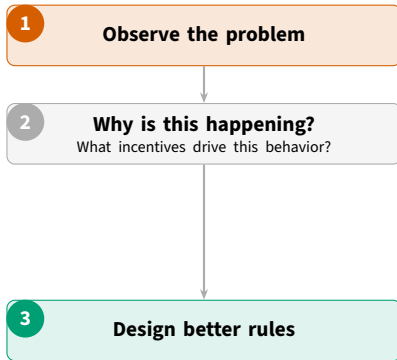
- Count the number of candies you took
- Your favorite candy is free
- Every additional piece costs \$5, paid to instructor in cash
- You can return any candy you don't want



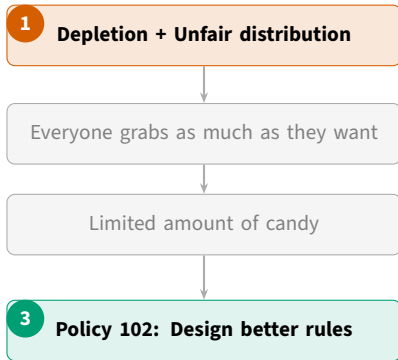
DEBRIEF: WHAT DID YOU NOTICE?

1. What happened to the candy under Yale Policy 101? Why?
2. Did Yale Policy 102 solve it? Why?

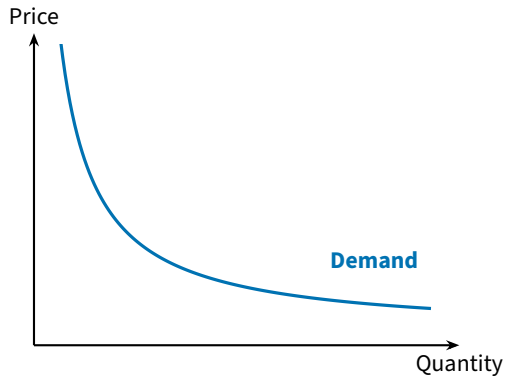
HOW ECONOMISTS THINK



THE PROBLEM

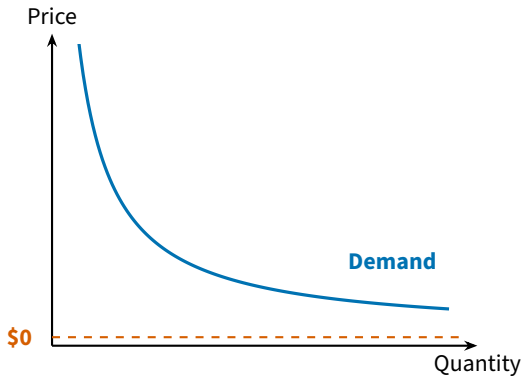


THE DEMAND CURVE



THE DEMAND FOR CANDY

Policy 101: Price = \$0



THE SUPPLY OF CANDY

The supply of candy is **finite**! Which means two things:

1. I cannot demand an infinite amount
2. Every unit I consume is one less unit for others!

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A good is rivalrous when one person's consumption reduces the amount available for others.

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Photo: Pexels (free license)

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Photo: Pexels (free license)

Finite goods $\implies$ Rivalrous goods

RIVALROUS GOODS

Rivalrous	Non-rivalrous
• Store clothing • Cars at a dealership	• Netflix movies • Toll road
• Ocean fish • Groundwater	• National defense • Public parks

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Question

Which items do you need to pay to use?

EXCLUDABLE GOODS

		Rivalrous	Non-rivalrous
Excludability	High	• Store clothing • Cars at a dealership	• Netflix movies • Toll road
	Low	• Ocean fish • Groundwater	• National defense • Public parks

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Excludable

A good is excludable when it's possible to prevent people who haven't paid from using it.

EXCLUDABLE GOODS

		Rivalrous	Non-rivalrous
Excludability	High	• Store clothing • Cars at a dealership	• Netflix movies • Toll road
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Question

What type of good is the candy in Yale Policy 101?

EXCLUDABLE GOODS

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Excludability	High	• Store clothing • Cars at a dealership	• Netflix movies • Toll road
	Low	• Ocean fish • Groundwater Yale candy	• National defense • Public parks

FOUR TYPES OF GOODS

		Rivalrous	Non-rivalrous
Excludability	High	Private Goods • Store clothing • Cars	Club Goods • Netflix • Toll road
	Low	Common Resources • Ocean fish • Groundwater • Yale candy	Public Goods • National defense • Public parks

TRAGEDY OF THE COMMONS

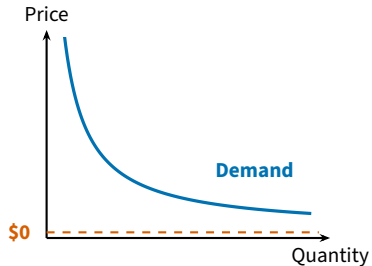
Tragedy of the Commons

When individuals deplete a common resource.

Why it happens:

- The good is **non-excludable** & demand is infinite
- The good is **rivalrous** because supply is finite
- The result is depletion

Price = \$0 on a common resource



TRAGEDY OF THE COMMONS

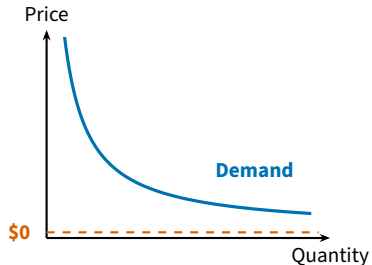
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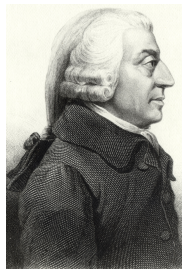
Question

Why can't people self-regulate their behavior?

SELF-INTEREST

“It is not from the benevolence of the butcher, the brewer, or the baker that we expect our dinner, but from their regard to their own interest.”

— Adam Smith, *The Wealth of Nations* (1776)

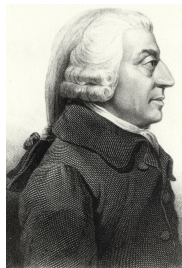


Adam Smith (1723–1790)

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What is Adam Smith saying?

- People act in their own interest to maximize their own happiness
- This is not greed — it is **rational behavior**
- He argues that, overall, it's actually good for society that everyone acts this way!

Nobody is being “bad,” we are acting in our self-interest. This can have both good and bad outcomes.

EXTERNALITIES

Externality

An **externality** is a cost or benefit that affects someone **who is not directly involved** in a decision or transaction.

Negative externality

- Carbon emissions from driving → everyone faces climate change

Positive externality

- A neighbor plants a beautiful garden → the whole street enjoys the view

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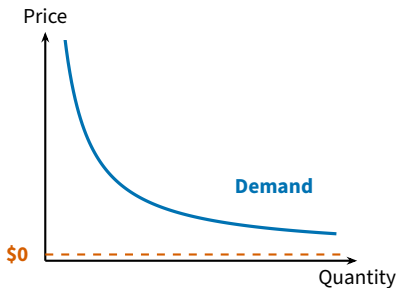
Which type of externality is the candy example?

HOW CAN WE FIX THE “TRAGEDY OF THE COMMONS”?

Tragedy of the Commons

When individuals deplete a common resource.

Price = \$0 (Yale Policy 101)

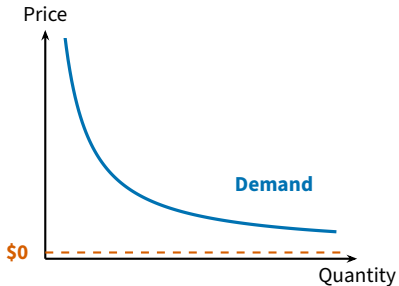


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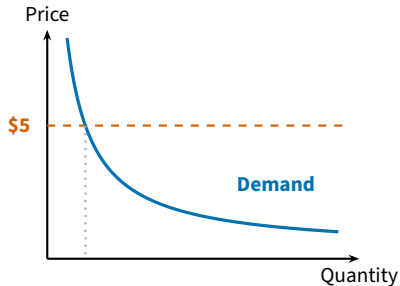
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Price = \$0 (Yale Policy 101)



Price = \$5 (Yale Policy 102)

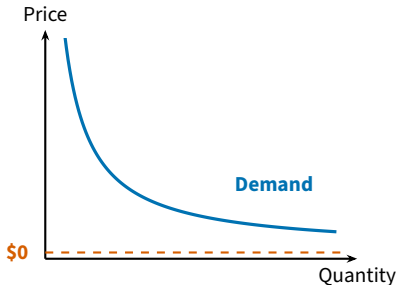


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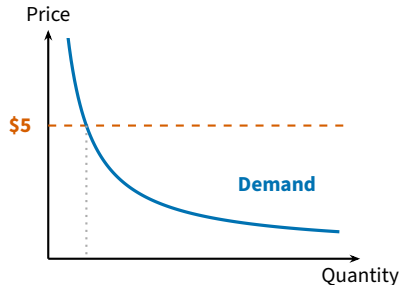
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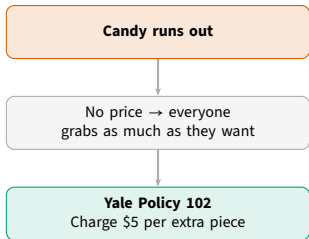
Price = \$5 (Yale Policy 102)



By raising the price, we reduce demand and prevent overuse. This is the logic behind taxes and fines.

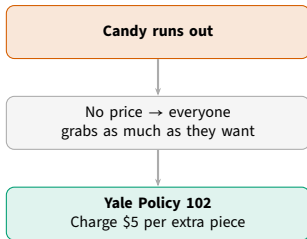
FROM CANDY TO CLIMATE CHANGE

The Candy Problem

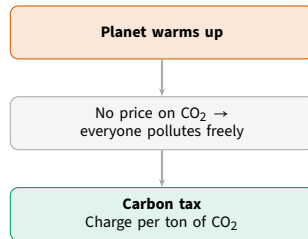


FROM CANDY TO CLIMATE CHANGE

The Candy Problem



Climate Change



Same logic, bigger scale: the atmosphere is a common resource.

OUTLINE

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2. Carbon Tax Policy Discussion

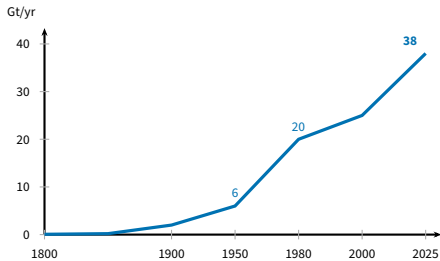
3. Recap

POLICY DISCUSSION

***Should the U.S. put
a price on carbon?***

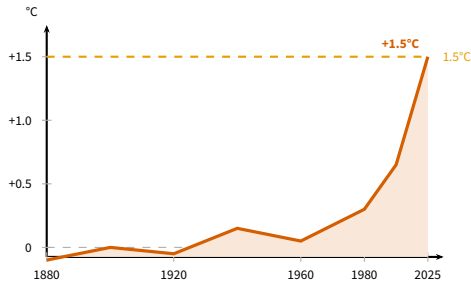
THE CAUSE AND THE CONSEQUENCE

Global CO₂ Emissions (Gt/yr)



Source: Global Carbon Project (2025).

Temperature Anomaly (°C vs. 1850–1900)



Source: NASA GISS; Copernicus.

Most emissions have occurred in the last 70 years. The Earth is now 1.5°C warmer — right at the Paris Agreement target.

WHO EMITS IN THE U.S.?

U.S. Greenhouse Gas Emissions by Sector (2022)

Sector	%	Households	Firms
Transportation	28%	Cars, personal trucks	Freight, airlines, shipping
Electric Power	25%	Home electricity use	Factory & office electricity
Industry	23%	—	Manufacturing, refining, chemicals
Comm. & Residential	13%	Home heating, refrigerants	Office & retail heating
Agriculture	10%	—	Livestock, crops, fertilizer

Source: U.S. EPA, *Inventory of U.S. Greenhouse Gas Emissions and Sinks: 1990–2022*.

A POLICY PROPOSAL: TAX CARBON EMISSIONS

Carbon Tax

A fixed price charged for every **ton of CO₂** emitted. The polluter pays for the damage their emissions cause.

How it works:

- Government sets a price, e.g. **\$50 per ton** of CO₂
- Firms and households pay when they emit

WHAT DOES \$50 PER TON OF CO₂ MEAN?

Answer

Gas price: \$3.50 → **\$3.94**

An increase of about **13%**

Source: EPA, "Greenhouse Gases Equivalencies Calculator."

Average U.S. gas price as of 2025.

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Answer

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Who is affected?



Firms



Households



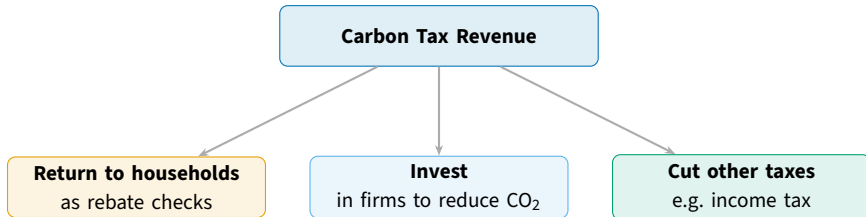
High income



Low income

WHAT SHOULD THE GOVERNMENT DO WITH THE REVENUE?

A carbon tax generates **billions of dollars** in revenue every year. The government has three main options:



REAL-WORLD EXAMPLE: CANADA

THE GLOBE AND MAIL
Toronto, ON 2024

**CANADA PRICES CARBON:
CA\$80 per Ton of CO₂**

90% of revenue returned
directly to households

[Placeholder text represented by horizontal lines]

What happened:

- Canada introduced a carbon tax of **CA\$80/ton** in 2024
- **90%** of revenue is returned to households via quarterly rebate checks
- **8 out of 10 households get back more** than they pay in higher prices

Source: Canada Parliamentary Budget Officer (2024).

POLICY DEBATE

Question

Economic analysis: Should the U.S. put a price on carbon?

Three stakeholder groups will analyze the question:

- Households
- Firms
- Government

Your Task: For your assigned group, answer:

1. What are the **benefits** of a carbon tax for your group?
2. What are the **costs** of a carbon tax for your group?
3. What is your **policy recommendation**?
Choose one of the following:
 - a. No tax
 - b. Tax + rebate to households
 - c. Tax + green investment
 - d. Other (formulate your own policy!)

STAKEHOLDER ANALYSIS: HOUSEHOLDS

Benefits

- Cleaner air → fewer health problems (asthma, heart disease)
- Rebate checks could exceed what most families pay in higher prices
- Drives innovation in affordable clean energy
- Protects future generations from climate damage

Costs

- Higher gas, electricity, and heating bills
- Poor households spend a larger share of income on energy
- Rural families with no transit alternatives are hit hardest
- Higher prices on everyday goods (food, shipping)

STAKEHOLDER ANALYSIS: FIRMS

Benefits

- Clear price signal helps long-term planning
- Early movers gain competitive advantage in clean technology
- Level playing field if all firms face the same carbon price
- New markets for low-carbon products and services

Costs

- Higher production costs (energy, raw materials)
- Disadvantage vs. foreign competitors without a carbon tax
- Energy-intensive industries may relocate abroad (“carbon leakage”)
- Short-run job losses in fossil fuel sectors (coal, oil, gas)

STAKEHOLDER ANALYSIS: GOVERNMENT

Benefits

- New revenue stream (\$250B/year at \$50/ton)
- Meets international climate commitments (Paris Agreement)
- Reduces future costs of climate disasters (floods, wildfires)
- Market-based approach — less bureaucracy than regulations

Costs

- Political backlash — voters dislike higher energy prices
- Administrative cost of implementing and enforcing the tax
- If the U.S. acts alone, global emissions may not fall much
- Revenue spent on rebates cannot fund other priorities

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TODAY'S KEY TAKEAWAYS

1. The **Tragedy of the Commons** arises when goods are **common resources**:
 - **Rivalrous**
 - **Non-excludable**
2. People demand too much and **deplete** the resource. This imposes a **negative externality** on others: they lose the opportunity to consume.
3. This problem can be mitigated with a **price system**, which changes incentives and generates revenue.
4. We applied this logic to **climate change**
5. But there are **always costs** to such policies! Different stakeholders (households, firms, government) are affected differently.